

BST Physics Curriculum Vol 10 Chapter 4 — PDEs + Heat / Wave / Laplace v0.4 (refilled per Cal #104)

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Vol 10 Chapter 4 — PDEs + Heat / Wave / Laplace

Chapter motivation

Standard graduate PDE methods: linear 2nd-order PDEs classified (elliptic + parabolic + hyperbolic); canonical equations (heat $\partial u / \partial t = \alpha \nabla^2 u$, wave $\partial^2 u / \partial t^2 = c^2 \nabla^2 u$, Laplace $\nabla^2 u = 0$, Poisson $\nabla^2 u = f$); separation of variables; Green's function method for inhomogeneous problems; boundary-value problems (Dirichlet + Neumann + mixed). Standard texts: Strauss + Evans + John.

BST cross-link: the heat kernel on D_{IV}^5 plays load-bearing role across BST framework — Vol 6 Ch 5 LOAD-BEARING identifies partition function $Z(\beta) = \text{Tr}(e^{-\beta \text{Casimir}})$ as heat kernel evaluation on Bergman $H^2(D_{IV}^5)$; Vol 4 Ch 4 T1485 Λ derivation IS heat-kernel at cosmological scale $t_{\text{cosmo}} = 47$. Heat-kernel chain Toy 273-639 verifies speaking-pair structure $k = 2, \dots, 20$ (19 consecutive levels).

Section 4.0b — Reader-grade 3-level pedagogy

Level 1: Standard PDE methods (heat + wave + Laplace + Poisson + Green's function); BST cross-link: heat kernel on Bergman $H^2(D_{IV}^5)$ = substrate partition function (Vol 6 Ch 5 LOAD-BEARING) + cosmological constant Λ (Vol 4 Ch 4 T1485) — heat-kernel framework underlies thermo + cosmology.

Level 2 (graduate-physicist): Standard PDE classification: linear 2nd-order PDE $A u_{xx} + 2B u_{xy} + C u_{yy} + \dots = 0$ classified by discriminant $\Delta = B^2 - AC$. Elliptic ($\Delta < 0$): Laplace $\nabla^2 u = 0$, Poisson $\nabla^2 u = f$; well-posed boundary-value problems with Dirichlet/Neumann conditions. Parabolic ($\Delta = 0$): heat equation $\partial u / \partial t = \alpha \nabla^2 u$; well-posed initial-value problem. Hyperbolic ($\Delta > 0$): wave equation $\partial^2 u / \partial t^2 = c^2 \nabla^2 u$; well-posed initial-value problem with characteristics. Separation of variables (Fourier series + Bessel functions); Green's function method: $G(x, x')$ satisfies

$L G = \delta(x - x')$; solution $u(x) = \int G(x, x') f(x') dx'$. BST substrate cross-link: heat kernel $K(t, z, z')$ on Bergman $H^2(D_{IV}^5)$ satisfies $(\partial/\partial t + \text{Casimir}) K = 0$ with $K(0, z, z') = \delta(z - z')$; per Vol 6 Ch 5 LOAD-BEARING $Z(\beta) = \text{Tr}(e^{\{-\beta \text{Casimir}\}}) = \int_{D_{IV}^5} K(\beta, z, z) dV(z)$ heat-kernel trace = partition function. Heat-kernel expansion $K(t) = \sum_k a_k(z) t^{(k-n/2)}$ (Seeley-DeWitt coefficients) with substrate speaking-pair structure period $n_C = 5$ verified $k = 2, \dots, 20$ (Toy 273-639 chain). Cross-link Vol 4 Ch 4 T1485 Λ derivation: same heat-kernel framework at cosmological scale $t_{\text{cosmo}} = g^2 - \text{rank} = 47$ gives $\Lambda = g \cdot \exp(-C_2 \cdot t_{\text{cosmo}}) = 7 \cdot \exp(-282)$ at 0.076 dex (99.9% of 122-order hierarchy). Wave + Laplace + Poisson substrate-cartography readings via substrate-tick $GF(128)^k$ discrete operators.

Level 3 (5th-grader accessible): Standard partial differential equations (PDEs) describe how quantities change in multiple variables (space + time). The heat equation describes how temperature spreads through a metal bar; the wave equation describes vibrating strings + EM waves; Laplace equation describes equilibrium states. BST identifies the heat kernel on the substrate D_{IV}^5 as the master function — Vol 6 Ch 5 LOAD-BEARING shows the partition function (thermodynamic master formula) IS the heat-kernel trace on Bergman H^2 ; Vol 4 Ch 4 shows the cosmological constant Λ IS the heat-kernel at large scale.

Section 4.1 — Standard PDE Classification

$A u_{xx} + 2B u_{xy} + C u_{yy} + \dots = 0$; discriminant $\Delta = B^2 - AC$ classifies as elliptic ($\Delta < 0$) / parabolic ($\Delta = 0$) / hyperbolic ($\Delta > 0$).

Canonical equations: - Heat (parabolic): $\partial u / \partial t = \alpha \nabla^2 u$ - Wave (hyperbolic): $\partial^2 u / \partial t^2 = c^2 \nabla^2 u$ - Laplace (elliptic): $\nabla^2 u = 0$ - Poisson: $\nabla^2 u = f$

Section 4.2 — Separation of Variables + Green's Functions

Separation: $u(x, y, \dots) = X(x) Y(y) \dots$ reduces PDE to system of ODEs (Sturm-Liouville per Vol 10 Ch 3).

Green's function: $G(x, x')$ satisfies $L G = \delta(x - x')$ with boundary conditions; solution $u(x) = \int G(x, x') f(x') dx'$ for inhomogeneous source f .

Section 4.3 — Substrate Heat Kernel on Bergman $H^2(D_{IV}^5)$

Heat kernel $K(t, z, z')$ on D_{IV}^5 satisfies $(\partial/\partial t + \text{Casimir}) K = 0$ with $K(0, z, z') = \delta(z - z')$.

Per Vol 6 Ch 5 LOAD-BEARING: $Z(\beta) = \text{Tr}(e^{\{-\beta \text{Casimir}\}}) = \int_{D_{IV}^5} K(\beta, z, z) dV(z)$ — heat-kernel trace IS partition function.

Heat-kernel expansion: $K(t) = \sum_k a_k(z) t^{(k - n/2)}$ with Seeley-DeWitt coefficients $a_k(z)$.

Section 4.4 — Heat-Kernel Chain Toy 273-639 (Speaking-Pair Structure)

Substrate speaking-pair structure with period $n_C = 5$ verified at heat-kernel levels $k = 2, 3, \dots, 20$ (19 consecutive levels, 4 full periods).

a_{16} confirmed (Toy 639): $\text{ratio}(k=16) = -24 = -(n_C^2 - 1) = -\dim \text{SU}(5)$ — uniqueness identity $n_C^2 - 1 = (n_C - 1)!$ holds ONLY at $n_C = 5$.

Section 4.5 — Cross-Link Vol 4 Ch 4 Λ (T1485)

Same heat-kernel framework at cosmological scale $t_{\text{cosmo}} = g^2 - \text{rank} = 47$:

$$\Lambda / M_{\text{Pl}}^4 = g \cdot \exp(-C_2 \cdot t_{\text{cosmo}}) = 7 \cdot \exp(-282)$$

$\log_{10}(\Lambda / M_{\text{Pl}}^4) = -121.63$ vs observed $-121.55 \rightarrow 0.076$ dex (99.9% of 122-order hierarchy resolved).

Section 4.6 — Wave + Laplace + Poisson Substrate-Cartography

Wave equation substrate-cartography: substrate-tick $\text{GF}(128)^k$ discrete operator framework + Bergman kernel propagation (T2457 Bergman = Feynman propagator structural-role-of, Friday).

Laplace $\nabla^2 u = 0$ substrate-cartography: harmonic functions on D_{IV}^5 + Bergman reproducing kernel.

Poisson $\nabla^2 u = f$ substrate-cartography: Green's function for substrate Casimir operator.

Section 4.7 — Honest scope + Connection

- Standard PDE methods + Green's functions [\[?\]](#)
- Heat-kernel BST cross-link Vol 6 Ch 5 LOAD-BEARING + Vol 4 Ch 4 [\[?\]](#)
- 19 consecutive heat-kernel speaking-pair levels verified $k=2..20$
- Open scope: explicit wave + Laplace + Poisson substrate-derivation (multi-week)

Connection: - Vol 6 Ch 5 LOAD-BEARING heat kernel + partition function - Vol 4 Ch 4 Λ from Substrate (T1485 + heat-kernel at cosmological scale) - Vol 10 Ch 3 ODEs + Sturm-Liouville (separation reduces to ODEs) - Vol 11 Ch 3 Wallach K-type (eigenfunction basis)

— Lyra, Vol 10 Ch 4 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT